

Adèle DEJOIE

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Introduction

I am a graduate student in the MVA Master's program at ENS Paris-Saclay, with research experience in LLM-based architectures for student modeling, probabilistic modeling, and stochastic control.

I am seeking a 6-month research internship starting in January 2027, with a preference for multimodal learning, NLP, representation learning, or model architectures.

Education

ENS Paris-Saclay – Paris-Saclay University Sept 2025 – Sept 2026
Master 2 in Research, MVA: Mathematics, Vision, Learning

- One year of advanced coursework in modern machine learning and its mathematical foundations, taught by leading experts in the field.
- Most courses include an independent research project carried out in pairs or small groups, fostering autonomy, collaboration, and scientific rigor.
- **Selected coursework:** Deep Learning; DL Theory; Deep Generative Models; Convex Optimization; ML for Text and Graph Data; Speech and NLP; Multimodal XAI; RL; Graphs in ML, ML for Time Series.

ENS Rennes & Magistère in Mathematics – University of Rennes Sept 2023 – July 2025
M1 in Fundamental Mathematics; BSc in Mathematics; BSc in Computer Science.

Lycée Henri-IV, Paris Sept 2020 – July 2023
Scientific Preparatory Classes

- Three years of intensive post-secondary education preparing for entrance exams to France's top engineering and research schools, with advanced coursework in Mathematics, Physics, and Computer Science.

Experience

Research Intern Bordeaux, France
INRIA - Flowers Team May 2026 – Nov 2026

- Research on Knowledge Tracing and student answer prediction under the supervision of **P.Y. Oudeyer** & **L. Gaven**. Designed architectures to efficiently inject structured histories into foundation models.
- Obtained promising preliminary results on answer prediction with a research article as a long-term target.

Mathematics Research Intern Osaka, Japan
Osaka University June 2025 – July 2025

- Introduction to stochastic calculus: Itô calculus, stochastic differential equations, generators, etc.
- Study of the optimal stopping problem. Control and optimization of stopping strategies.

Research Intern Lille, France
INRIA June 2024 – July 2024

- Designed, generated, and refined Bayesian networks to create valid probabilistic problems under minimal assumptions. Used the generated dataset to fine-tune a BERT-based language model.

Technical Skills

Research areas: Deep learning, LLMs, representation learning, generative models, NLP, multimodal learning.

Tools: PyTorch, Hugging Face Transformers, PEFT/LoRA, NumPy, pandas, SciPy, Matplotlib, Git, LaTeX.

Programming: Python, Java, C, C++, SQL. **Languages:** French native, English fluent, Japanese (learning).

Selected Projects

Molecule-to-Text Generation: generated natural-language molecular descriptions from molecular representations; sequence generation and evaluation.

Deep Generative Models for Graph Data: implemented VAE-style generative models for graph-structured data and network generation.

Reports

Dejoie, A. (2025). *Study of the Optimal Stopping Problem using Stochastic Calculus*, University of Osaka – Presented at ENS Rennes.

Dejoie, A. (2024). *Generation and Fine-Tuning of Probabilistic Problem Sets using Bayesian Graphs*, INRIA Lille – Presented at ENS Rennes.